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Design for Greenfield Local Hub

Functional and Non-functional Requirements (FR and NFR)

FR

1 Customers:

1.1 can see the price of an item

1.2 add items to cart and modify quantity

1.3 choose delivery or location options along with available time slots

1.4 schedule recurring orders

1.5 choose a delivery location

1.6 cancel orders within an allowed window

1.7 view order history

2 Producers:

2.1 create, edit and remove products

2.2 manage stock levels

2.3 view incoming orders

2.4 set or adjust prices

3 Admin

3.1 create/manage user accounts, and assign roles and permissions to accounts

3.2 manage and reschedule orders

3.3 create/manage loyalty programs

3.4 log audit trails

4 System

4.1 users can register and log in

4.2 passwords are processed securely using encryption

4.3 response times should be kept under a second

4.4 provide accessible UI compatible with WCAG

4.5 navigation bar will be displayed to all users

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NFR

1 Performance

1.1 website will load pages within 3 seconds

1.2 website will support multiple users at a time

2 security

2.1 user data will be stored securely

2.2 the system will use a secure login authentication

2.3 the website will comply with UK GDPA and DPA 2018

3 usability

3.1 the website will be easy to navigate and user-friendly

3.2 product info will be clear and easy to understand

3.3 users should be able to complete orders with minimal steps

4 accessibility

4.1 the website will be accessible to users with disabilities

4.2 it will follow accessibility guidelines (good colour contrast and screen reader compatibility)

4.3 compliance with the equality act 2010

5 reliability

5.1 available most of the time with minimal downtime

5.2 orders and data should not be lost if an error occurs

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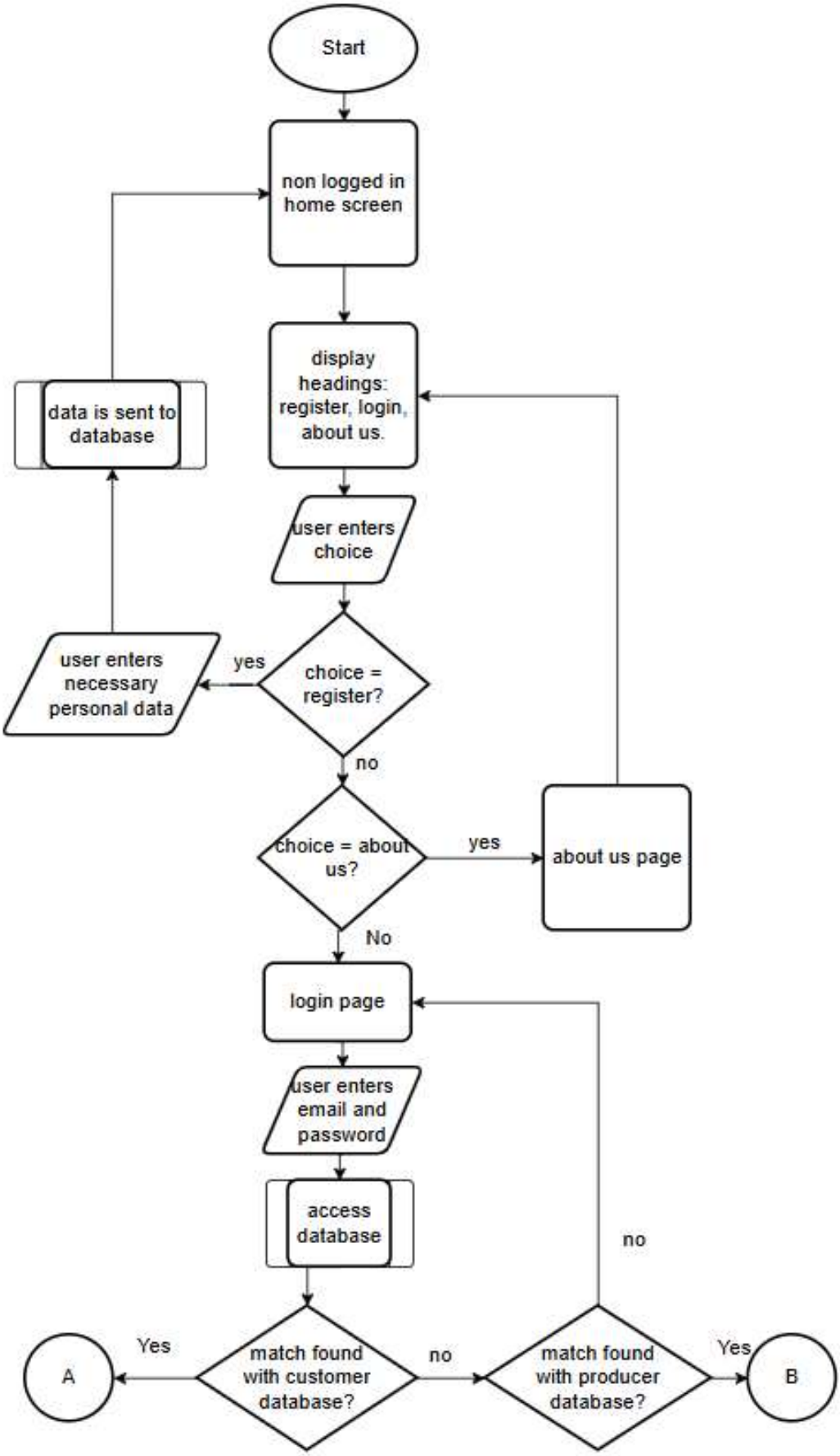
Decomposition



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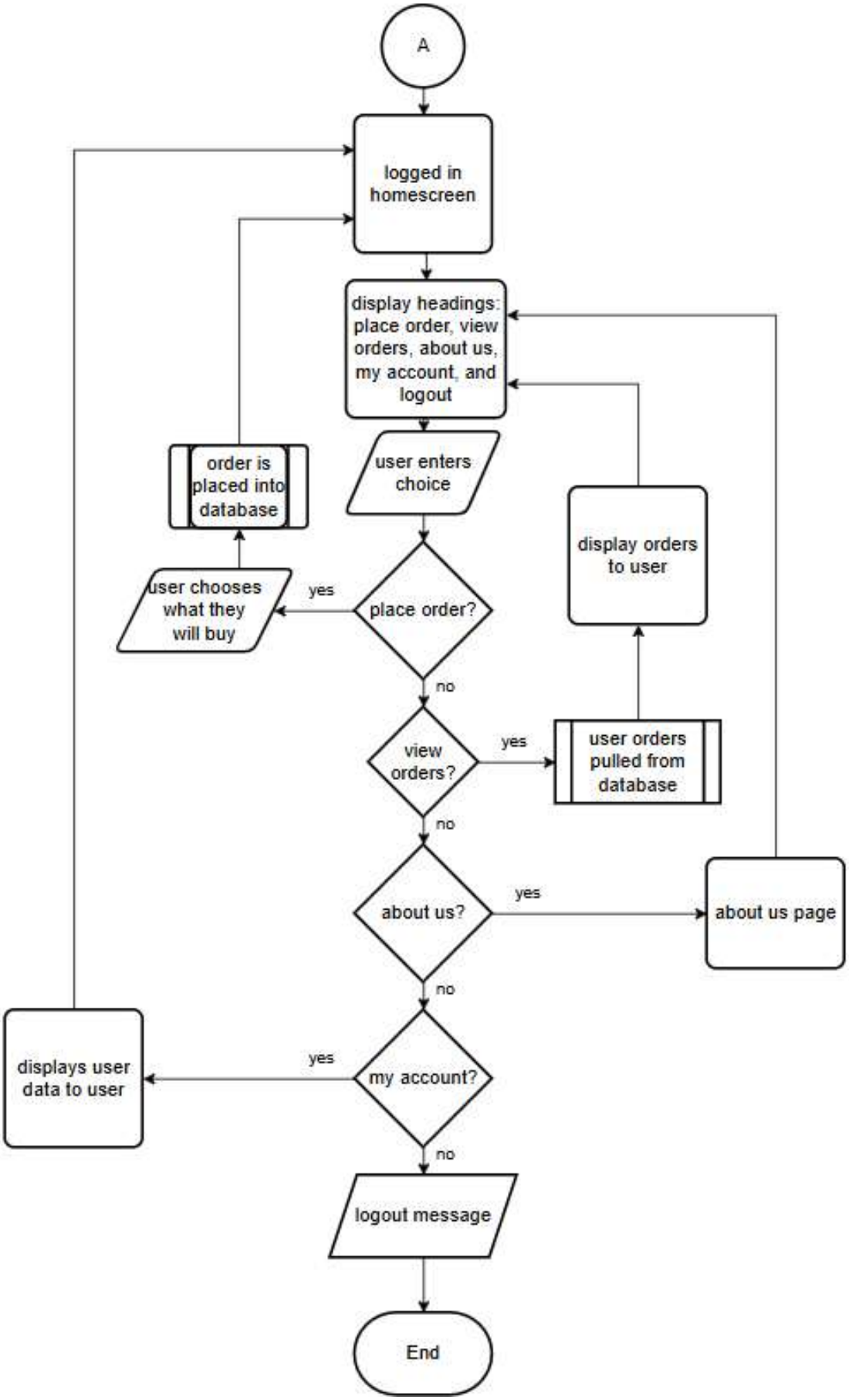

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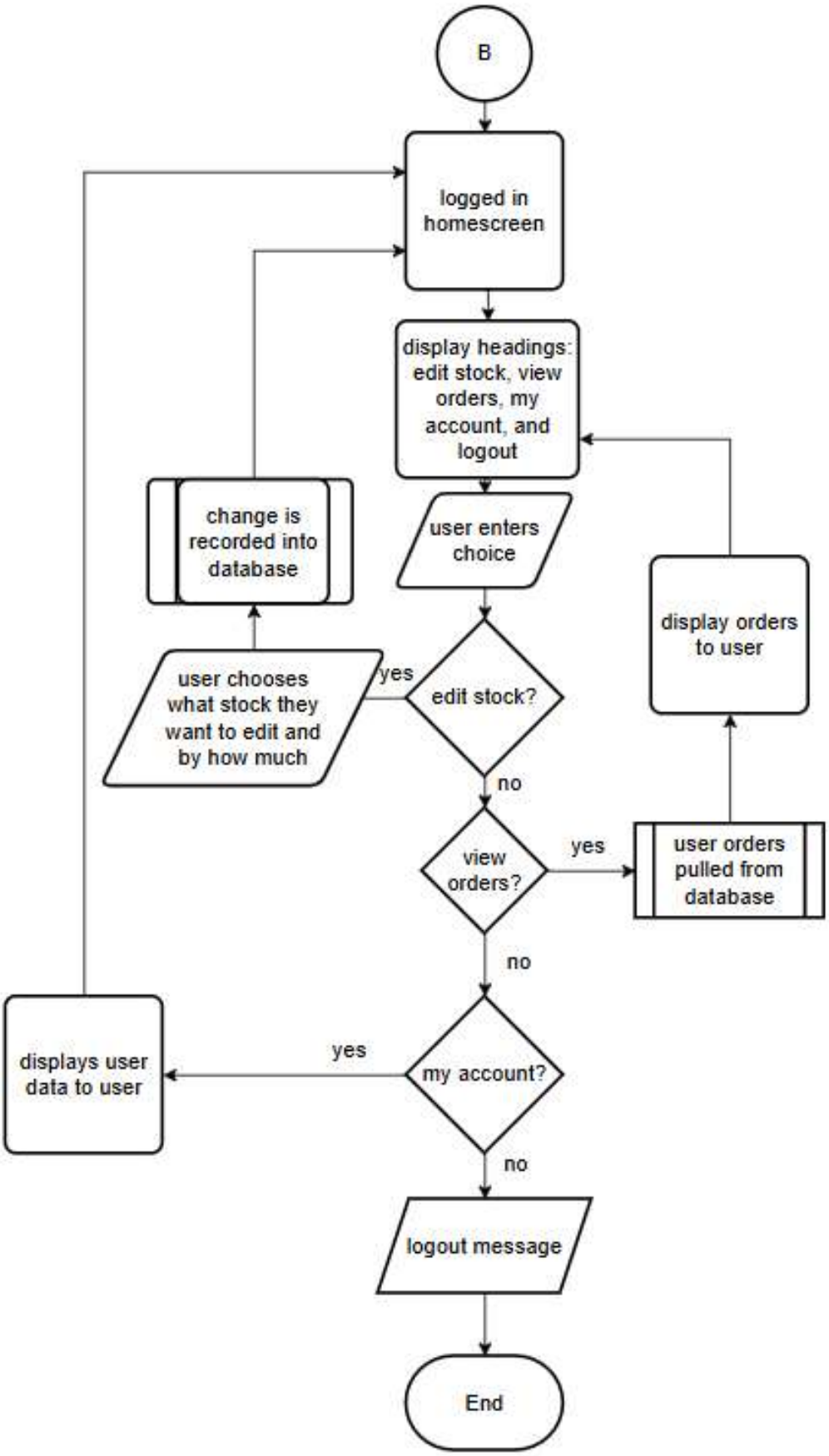

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Decomposition Justifications

- user will start non logged in to begin with so therefore they will have a different homepage to logged in users there are some website functions that require the user to log in.
- the navigation bar will then display all the options they can choose from, fulfilling FR 4.5. hopefully these options will be clear and concise, partially fulfilling NFR 3.1
- the user can then click on a section they want to choose, this said page should load quick enough to meet NFR 1.1
- If the user chose register, then they will enter in their details depending if they are registering as a customer or producer and then this data will be stored in the database - FR 4.1, and the password will be encrypted using sha256 – FR 4.2
- If the user chose about us, then the website will provide information about the producers, their methods and the benefits of buying locally.
- If the user didn’t chose either of these two then they chose login as that it the only option left. The user will enter their email and password, that is on the system, and then they will be logged on.
    - If they log on as a customer then their new headings will be: place order, view orders, about us, my account and logout
    - If the user logs on as a producer their new headings will be: edit stock, view orders, my account, logout
- these user will still click on the navbar to go to the intended page and that page will perform its intended function
- the flowcharts will only end when the user has chosen to logout.

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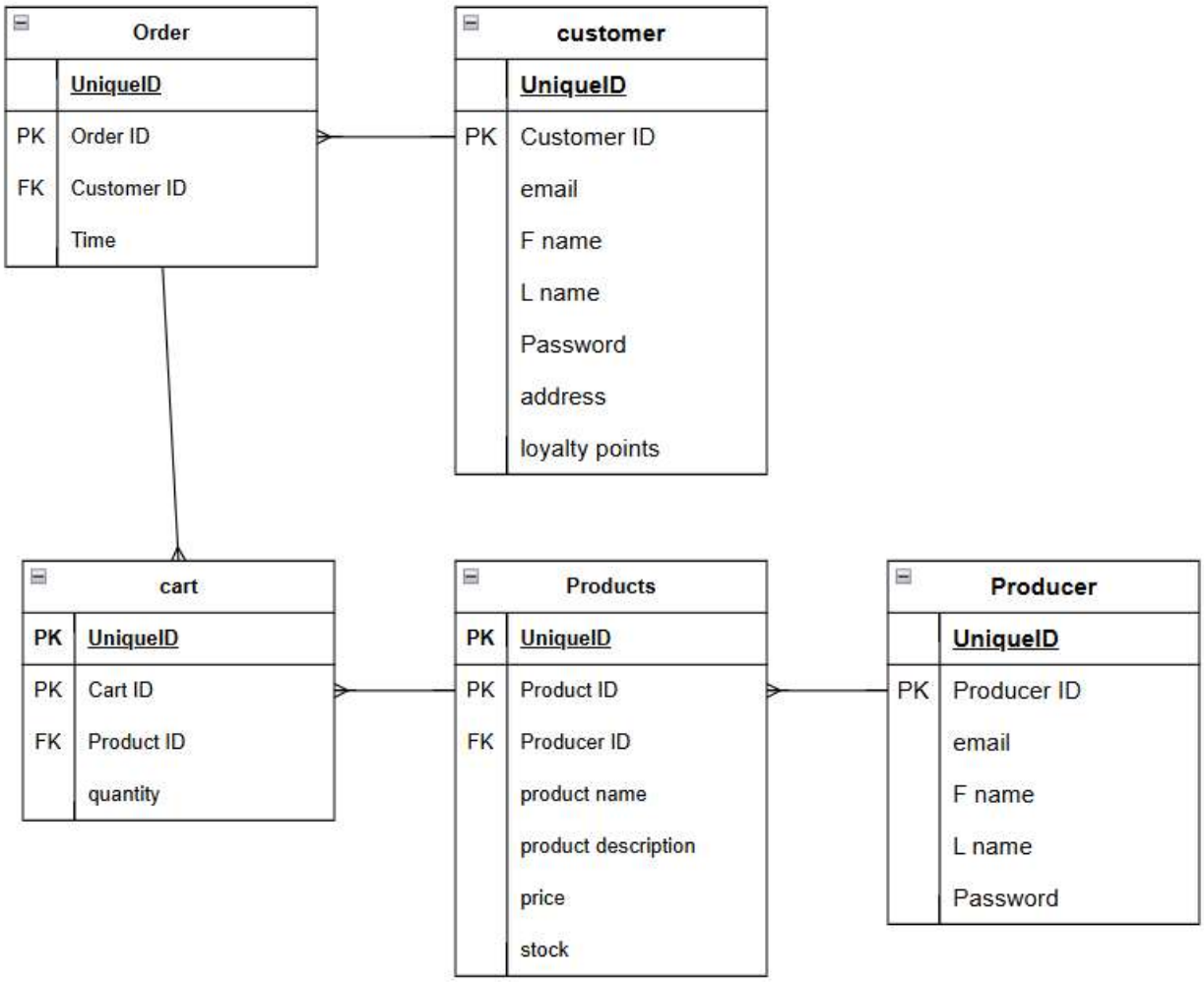
Entity Relationship (ER) Diagram



this is what the ER diagram looks like.

Justifications:

- A customer can have many orders but an order can only have one customer so it is a one to many relationship.
- And order can have many products and products can have many orders so I have created a cart entity that will co
- The order table has customer ID as a foreign key so that we can find both the customer easily from the order.
- a producer can have many orders but an order can only have one producer, so it is also a one to many relationship.

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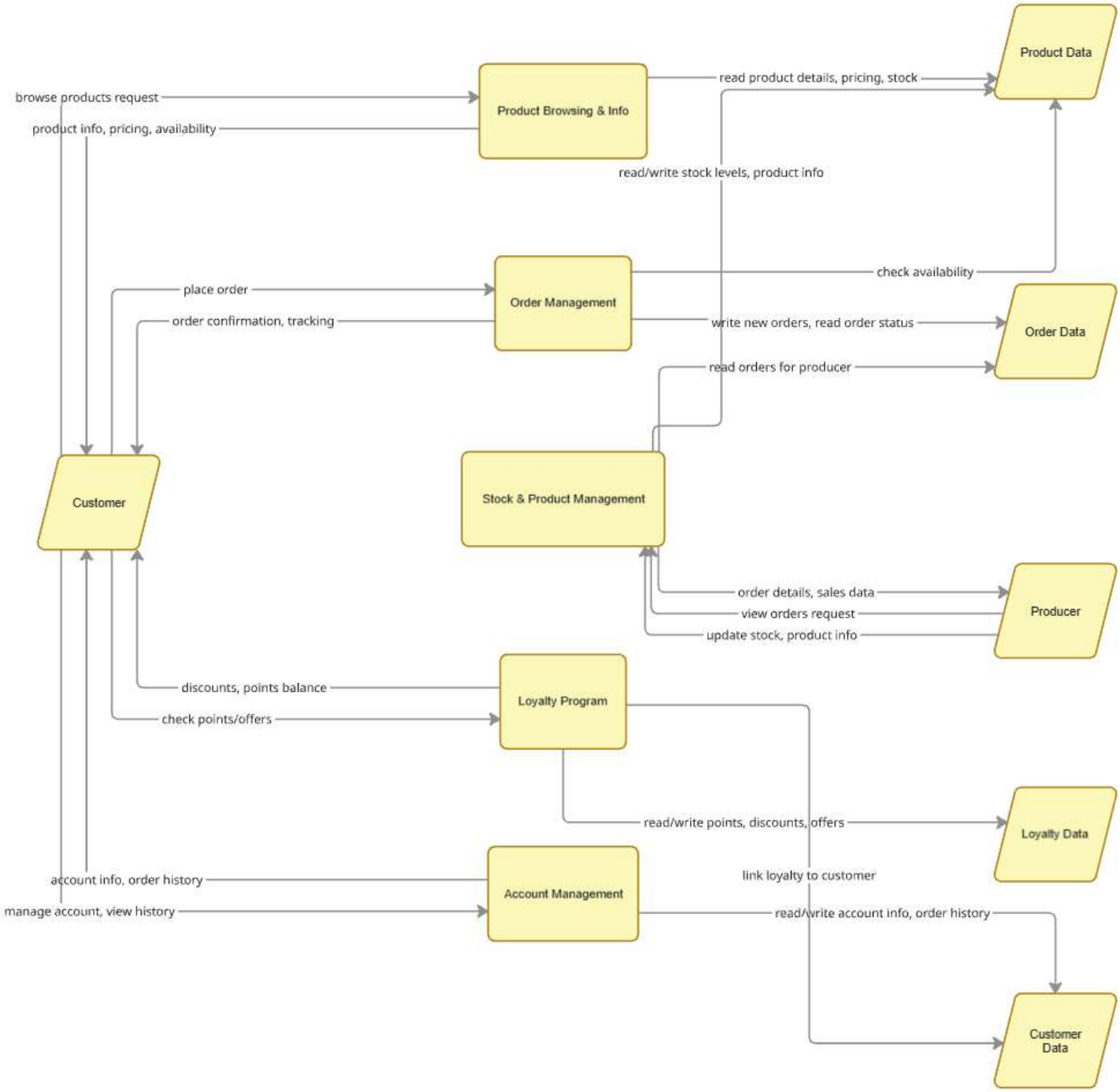
- Each entity has been assigned an ID so that the rest of the information will be easy to find if we know the ID.

Data Flow diagram



Justifications:

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Processes

Product browsing and information

- allows customers to view detailed product information, like pricing, availability etc
- essential for transparency and informed purchasing, aligning with GLH's idea of being open about producers and their products
- will require the product data in order to receive the information, will then provide customers with the product information on requesting

Order Management

- required to allow customers to place orders for collection or delivery.
- Reads and writes orders from order data
- will send order confirmation back to customers
- checks product availability from product data to validate orders

Stock and product management

- allows producers to update product details and stock levels and to view orders
- provides producer dashboard that is required for managing inventory and sales
- reads the order data that producers will have to supply
- updates the stock in product data
- recieves requests from producers to view orders and update stock/product information

loyalty program

- manages loyalty points, discounts and special offers
- is the implementation of the loyalty scheme GLH has asked for
- reads and writes loyalty points and discounts from loyalty data

Data stores

Product data – stores product descriptions, pricing, and stock levels, used both by producers and customers

Order data – holds details of placed orders, when they were placed for, whether they are delivery or pickup. These are also used by producer and customers

Customer data – contains customer accounts and their order history, used by account management and loyalty data.

Loyalty data – stores loyalty points and discounts, enabling the loyalty program to function

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User Summary

customers send browse request to receive product info, place orders and receive confirmation, check loyalty points and discounts, and manage account information and order history

producers update stock and product info, view order details and receive sales through the dashboard

Data dictionary

Variable name	Data type	Description	Example data	Reasoning
Customer_ID	integer	Unique identifier for each customer account	4	Allows system to identify each customer easily
Producer_ID	Integer	Unique identifier for each producer account	4	Allows system to identify each producer easily
F_name	String	First name of user	Bob	To personalise each account
S_name	String	Last name of user	Smith	To personalise each account
Email	String	User's email address	bobsmith@gmail.com	Used for account login
Password	string	User's password, will be stored using sha 256 to keep it secure and comply with FR 4.2	Passw0rd!	Adds security to account login
Address	String	Default delivery address	10 Market Street, Leeds	Required for giving orders to customers
Loyalty points	Integer	Points gained from loyalty system	50	Required for the loyalty rewards system to work
Product ID	Integer	Unique identifier for each product	4	Allows system to identify product
Product name	String	Name of product	Eggs	Allows customers to

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				identify product
Product description	String	Description of the product	The eggs are from free range chickens	Provides customer with detailed product info
Price	Float	Price of product	2.50	Display the price to customers
Stock	Integer	Quantity of product available	20	Allows system to track product availability
Order ID	Integer	Order identifier	4	Allows orders to be tracked
Quantity	Integer	Quantity of item ordered	1	To show how many units of product were purchased

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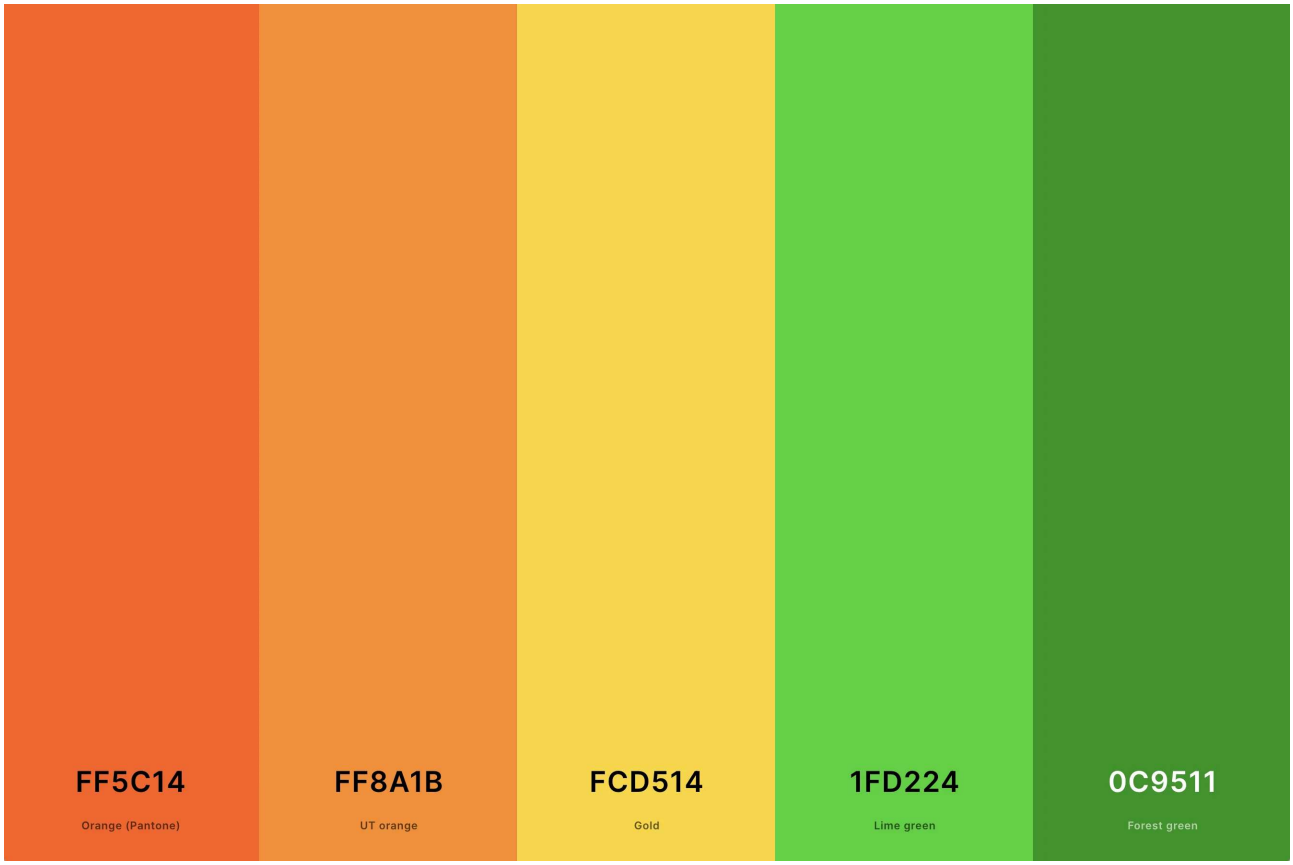
Colour scheme



These are the colours I will be using, I have chosen this palette as the colours are contrasting, making reading much easier and meeting the criteria of NFR 4.2. The green is associated with nature, farming, freshness and sustainable. Since GLH is a cooperative of local farmers and food producers, green should reinforce the idea of local, natural and sustainable food. The orange makes up a good secondary colour as it represents friendliness and community, these reinforce the idea of supporting local producers and community based shopping. It also, like green, gives food imagery, further supporting the ideas of GLH.

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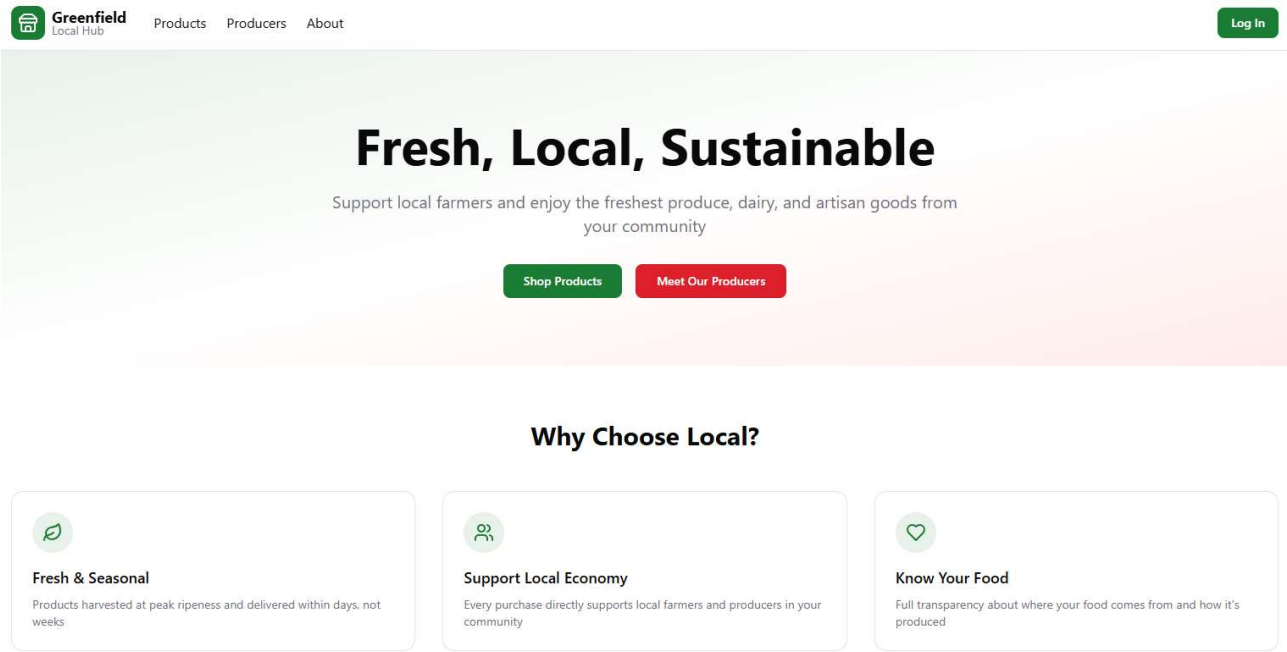
Interface design



Justifications:

I have chosen to make my buttons coloured so that users can identify them

The buttons are coloured green and red due to my colour scheme being primarily consistent of those two colours.

The navbar is at the top so that users can navigate easily but also explore the page they are on without inconvience

Pseudocode

Connect to database

```
FUNCTION connectToDatabase(connectionParameters)
    TRY
        connection = openDatabaseConnection(connectionParameters)
        IF connection IS successful THEN
            RETURN connection
        ELSE
```

```
                THROW "ConnectionFailedError"
        CATCH error
                DISPLAY "Database connection error: " + error.message
                RETURN null
END FUNCTION
```

Insert into database

```
FUNCTION insertData(connection, tableName, data)
        TRY
                query = buildInsertQuery(tableName, data) // e.g., "INSERT INTO tableName
                (fields) VALUES (values)"
                result = executeQuery(connection, query)
                IF result IS success THEN
                        RETURN true
                ELSE
                        THROW "InsertFailedError"
        CATCH error
                DISPLAY "Insert error: " + error.message
                RETURN false
END FUNCTION
```

Get data into database

```
FUNCTION getData(connection, tableName, filterCriteria)
        TRY
                query = buildSelectQuery(tableName, filterCriteria) // e.g., "SELECT * FROM
                tableName WHERE conditions"
                results = executeQuery(connection, query)
                RETURN results // Could be a list of records or empty if none found
        CATCH error
                DISPLAY "Data retrieval error: " + error.message
                RETURN null
END FUNCTION
```

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Process data from the database

```
FUNCTION processDataFromDatabase(data)
    IF data IS null OR data IS empty THEN
        DISPLAY "No data to process"
        RETURN null
    END IF

    processedData = initializeEmptyStructure()

    FOR EACH record IN data DO
        processedRecord = processSingleRecord(record) // e.g., calculate fields, format
        values
        ADD processedRecord TO processedData
    END FOR

    RETURN processedData
END FUNCTION
```

process data from input

```
FUNCTION processDataFromInput(rawInput)
    cleanedInput = validateAndSanitize(rawInput)

    IF cleanedInput IS invalid THEN
        DISPLAY "Invalid input data"
    RETURN null
    END IF

    processedInput = transformInput(cleanedInput) // e.g., parsing, data type conversions,
    enrichment

    RETURN processedInput
END FUNCTION
```

test strategy

This part of the design document is a separate file due to formatting issues